

ORIGINAL ARTICLE

Enhancing aviation safety and efficiency: applying artificial intelligence (AI) to address navigational challenges

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Abstract

Precision Navigation and Timing systems, integral to air traffic management, primarily rely on the Global Navigation Satellite System (GNSS) for accurate data transmission. However, GNSS signals, particularly those used in Automatic Dependent Surveillance-Broadcast (ADS-B) systems, are susceptible to interference, leading to potential safety risks in aviation. This study presents a novel AI-driven Flight Trajectory Prediction Framework, leveraging machine learning and deep learning techniques to detect and mitigate GNSS signal interference. By training models on ADS-B data, this framework identifies potential interference and evaluates its impact on air traffic. The implementation of this AI-based system enhances the reliability and security of air navigation, significantly reducing human error and elevating overall safety standards in aviation. Experimental results demonstrate the framework's efficacy in improving navigational accuracy and operational efficiency within modern air traffic control systems.

Keywords Security and safety · Machine learning · Deep learning · Global navigation satellite system (GNSS) · Jamming or message hijacking

1 Introduction

The use of Automatic Dependent Surveillance-Broadcast (ADS-B) systems has greatly enhanced the field of air traffic control. Technology is an integral part of modern-day airport infrastructure, thus improving the accuracy and timeliness of air traffic information through unencrypted broadcasting. Yet, in addition to the benefits of this technology, the ADS-B system is also susceptible to some serious threats to security. Particular areas of vulnerability, such as jamming of the communications channel and message hijacking, whereby the ADS-B message is taken over and manipulated without authority, pose serious threats. Such vulnerabilities are capable of significantly reducing the integrity of the system, thus exposing pilots and passengers to safety threats.

The rising trend of ADS-B technology in aviation has generated concerns in relation to potential network vulnerabilities in terms of security. The potential vulnerabilities in the ADS-B network could undermine the safety and integrity of the air transport system; thus, for the sake of proactive safety assurance and protection of ADS-B networks [1], there is the need to discover and remedy the threats to its security [2–4]. The goal is attainable through the adoption of an upper layer approach of strong authentication and encryption to secure means of communication, alongside the physical layer approach of creating protection to avoid physical attacks and improper access to the physical components of the ADS-B network [1].



In addition, Global Navigation Satellite Systems (GNSS), such as GPS and GLONASS, are essential to the functionality of Automatic Dependent Surveillance–Broadcast (ADS-B) by providing essential global positioning services. These systems, however, are susceptible to security risks such as jamming and spoofing attacks. The European Union and China are pursuing collaborative efforts in the development of new Global Navigation Satellite Systems (GNSS) with the aim of augmenting regional and global navigation operations. These efforts are designed to improve interoperability while at the same time mitigating security weaknesses.

The ADS-B system is susceptible to multiple attacks such as message spoofing, network security breaches, and communications disruption [7]. Technical solutions to counter these threats propose monitoring the timing of the transmission of the receiver and the sender to recognize forged ADS-B messages. Additionally, since GPS signals might be disrupted to lead to erroneous location information, ADS-B information must be authenticated [8]. Owing to the increasing menace of cyberattacks, there is the urgent need to implement an end-to-end approach to secure the safety and integrity of important infrastructures like ADS-B [9]. The proliferation of criminal activities and clandestine national cyber operations has raised the threats posed by SCADA and control systems, particularly used in managing air traffic [10].

Machine learning, deep learning, and artificial intelligence are crucial in solving these issues. This paper proposes a novel safety system based on various artificial intelligence techniques to predict and minimize risks in aerospace systems. It seeks to enhance safety procedures and maintenance by studying vast amounts of data in the Aviation Safety Reporting System (ASRS). Machine learning tools employed in maintenance predict the failing of equipment and schedule repairs by examining time-series data and event logs. It ensures smooth and consistent aircraft operations. Applying artificial intelligence in aerospace enhances the predictability of threats and improves the strength of the system, which enhances safety and efficiency in the aerospace industry.

Most of the studies considered the security issue of the Automated Dependent Surveillance–Broadcast (ADS-B) system, and fewer studies on artificial intelligence (AI) concerns. The central objective of the present study is to propose a method to utilize artificial intelligence to enhance ADS-B security. It applies sophisticated AI methods to discover, verify, and eliminate potential security threats. It enhances ADS-B safety, which is an integral component of contemporary flight safety, to maintain the trust and confidence of the public.

Classic approaches to flight-path forecasting rely significantly on fixed models or basic rules. Such approaches typically fail easily in crowded or dynamic airspace. They have difficulty dealing with issues such as noise or signal reflections, are not easily adaptable to abrupt changes in flight plans and are not very useful in forecasting atypical occurrences or flight-path changes. Most existing approaches do not handle time and context information effectively, hence are of limited use for practical purposes. With AI, particularly deep learning models, we can analyze sophisticated relationships across flight data in time and space. Yet most AI work thus far does not work in real-time, is difficult to interpret, or does poorly when the signal is weak.

This paper introduces AFTPF, an AI-based flight-path prediction system. The system is designed to blend various GNSS signals, adapt to real-time environments, and enhance aviation safety by detecting issues in advance. This article would like to bridge existing technology gaps with adaptable neural networks in an understanding model of the signal.

The structure of this study is outlined as follows: Sect. 2 provides an overview of the background information and basic concepts. Section 3 focuses on the discussion of related works. The proposed framework is presented in Sect. 4. The experimental results are analyzed and explained in Sect. 5. Section 6 concludes the study and discusses future research directions.

2 Background and basic concepts

Section 2.1 provides a brief overview of Automatic Dependent Surveillance–Broadcast (ADS-B). Next, we will focus on the integration of data with ADS-B. Section 2.3 focuses on Satellite Systems and Global Navigation Satellite Systems. Lastly, the impact of machine learning and deep learning was discussed in the field of aviation.

2.1 Automatic dependent surveillance-broadcast (ADS-B)

The Automatic Dependent Surveillance-Broadcast (ADS-B) system has emerged as an essential tool in air traffic control, experiencing an increasing adoption in aircraft. This technology, renowned for its openly accessible and freely available message data format, enhances the transparency and availability of air traffic information. The system utilizes the 1090ES, a sophisticated SSR Mode-S surveillance device, to integrate its data with traditional Mode-S surveillance. The integration allows for the concurrent operation of ADS-B and Mode-S transceivers [11].

Figure 1 illustrates the format of a 1090ES ADS-B message and the organization of the data link. The figure outlines data flow from GNSS satellite signals to GPS reception, then to onboard aircraft systems, and finally relaying to ADS-B ground stations. The major elements included are the reception of GPS signals, encoding modes (Mode A/C/S), ADS-B transmitter types (1090ES and UAT), and a schematic of the ADS-B message format showing the 112-bit frame structure including preamble, data block, and the individual message fields (DF, CA, AA, ME, PI). The diagram has been revised for better clarity, label readability, and publication-grade resolution.

Aside from the possibility of attacks on the ADS-B system at the network layer of the traffic management system, it is important to additionally acknowledge the potential for assaults on the GNSS signal, which is responsible for providing input for ADS-B location data. Ensuring the integrity of the GNSS signal is of utmost importance. It is vital to guarantee the precision and reliability of GNSS signals, navigation data, information, and air traffic management.

The potential threats of an ADS-B system are explored in [12]. Disregarding the potential threats to the ADS-B system jeopardizes flight traffic safety measures and risks the safety of pilots and passengers. The present study [13] attempts to introduce an alternate method for detecting potential threats to the ADS-B system. The method would encompass the monitoring of abnormal data and analysis of changes in the traffic patterns of the system. Such changes could indicate the possibility of an attack.

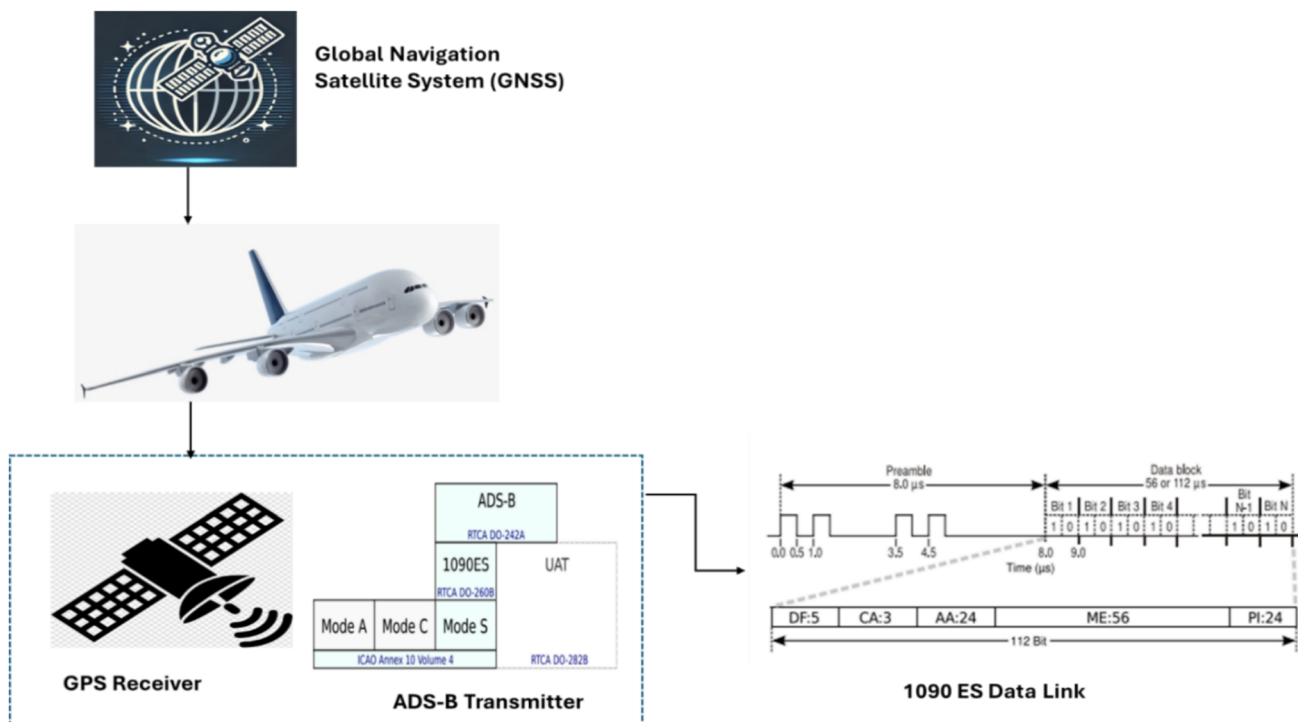


Fig. 1 The 1090ES ADS-B message format and the data link structure

The use of artificial intelligence for the detection of possible interferences of GNSS signals in the vicinity of an ADS-B site can considerably reduce the risk of such interferences affecting the ADS-B system. However, using deep learning and machine intelligence, an ADS-B system can improve its ability to prevent the risk of disruption of GNSS signals through awareness of their closeness to the ADS-B station.

2.2 Data and ADS-B integration

The Automatic Dependent Surveillance-Broadcast (ADS-B) system is crucial in maintaining communications between aircraft and ground facilities, supplying systematic information related to aircraft movement, signals, and sensor data. The technology is significant in improving the accuracy and reliability of air traffic control while protecting the system against jamming and message eavesdropping in the modern age of aerospace infrastructures. The ADS-B technology accommodates two major types of connections: (a) between aircraft and (b) between aircraft and ground facilities. The information disseminated by ADS-B covers extensive information related to aircraft, signals, sensors, and relevant factors. ADS-B technology broadcasts information encompassing dynamic and static parameters. The static information includes data related to the departure and arrival airports of an aircraft, its name, and the scheduled air routes along which it is meant to travel. On the other hand, the dynamic information contains 4D parameters, which encompass timestamps, geographic positions (latitude, longitude, and altitude), along with velocity and direction of heading [18].

2.3 Satellite systems and global navigation satellite systems

Global Navigation Satellite Systems (GNSS) are sets of satellites which emit signals. The signals are accessible anywhere in the world, regardless of the weather and time of day. Currently in 2010, there are two satellite navigation systems operational: the Global Positioning System (GPS), operated by the US Department of Defense, and the Global orbiting Navigation System (GLONASS), operated by Russia. The European Union and China are currently developing their own GNSS. The European Union's is called Galileo, and the Chinese is referred to as Compass. The systems are designed to function independently, in combination, and simultaneously, with the Chinese system to be extended to a wider area outside of its borders.

At present, the GPS network comprises of 31 operational satellites, which are uniformly distributed among six circular orbits. Each orbit consists of a minimum of five satellites. The positions of these orbits are separated by an angle of 60° and have an inclination of 55° with respect to the equator. The satellites move in approximately circular orbits that are not geostationary, and they complete one orbit every 11 h and 58 min. Generally, most locations on Earth should always have a minimum of three GPS satellites visible, but a minimum of four satellites is required for precise and continuous determination of a location on the globe [14].

The paper examines the functionalities of both operational and developmental GNSS systems, highlighting their ability to provide worldwide reception in all circumstances. The main goals of these systems are to achieve interoperability, independence, and complementarity. It is anticipated that in the future, most airplane will utilize multi-constellation receivers to analyze signals from different Global Navigation Satellite Systems (GNSS) in order to improve and supplement their performance. In [15], author provides a thorough examination of the functioning of these systems, as well as in-depth talks on certain existing difficulties.

2.4 Machine learning and deep learning in aviation

Despite the presence of numerous challenges, the extensive accessibility of abundant data and computer resources has accelerated the advancement and use of artificial intelligence methods across various industries. This research introduces a distinctive safety assurance framework for aviation systems. The model improves the accuracy of predicting risks and their severity in aviation systems by employing a combination of machine learning and deep learning techniques. The collective performance is fine-tuned by utilizing the diversity score of the fundamental

classifiers. The model is trained and assessed using 12 years' worth of data from the Aviation Safety Reporting System (ASRS) database. In order to demonstrate the superiority of our model, we employ precision, recall, and F-Score as quantitative evaluation metrics [16].

The application of machine learning serves the twin purposes of enhancing safety and predictive maintenance. Predictive maintenance has in recent years grown in importance in numerous industries due to its ability to provide forecasts of hardware failures and contribute to proactive scheduling of maintenance. This kind of maintenance is realized through automated data analysis methods that continuously monitor the status of hardware components during their working life, usually presented in the form of time-series data and event logs. An overriding challenge in this regard is the analysis of large amounts of historical data in order to create advanced predictive models that can effectively predict upcoming failures. The model leverages regression-based methods in combination with machine learning approaches, including Gaussian mixture models, for the assessment of risk levels. The strategy proves effective as a scalable solution for deployment within clustered environments, especially regarding training and evaluation processes [17].

2.5 Gated recurrent units (GRUs)

Gated Recurrent Units (GRUs) are one of the kinds of Recurrent Neural Networks (RNN) designed to address vanishing gradient issues. Such issues occur when it is difficult to train normal RNNs on long sequences of data. GRUs simplify the process by having two gates—reset and update—to determine how data flows. This assists the model to decide what data to retain or discard. Due to this, GRUs can learn and remember patterns in data as time passes and are suitable for such tasks as time-series analysis and detecting anomalies in data.

In aviation, GRUs are highly applicable to examining flight data, where details, such as speed and altitude, are relevant to functions such as flight-path prediction and issue identification. Their capacity to process sequences of varying lengths makes them ideal for real-time systems. Studies have determined that GRUs outperform standard RNNs in accuracy and efficiency. For instance, research has determined that GRUs enhance 4D trajectory prediction through the comprehension of intricate relationships in ADS-B data with higher accuracy and reliability compared to conventional approaches. They have also been beneficial for forecasting maintenance requirements through the analysis of patterns in sensor data to predict equipment failures [25, 26].

GRUs are crucial in systems such as the AI-based Flight Trajectory Prediction Framework (AFTPF), in which they can adapt and function well. GRUs enhance safety in air traffic management.

Currently, Gated Recurrent Units (GRUs) are applied to indicate time relationships since they are easy and able to identify short- to medium-term patterns rapidly. Nevertheless, GRUs tend to struggle to maintain long-term context in longer sequences. To rectify this, subsequent versions of AFTPF may employ the use of Continuous-time Augmented Recurrent Unit (CARU) [30]. This unit assists with memory since it constantly updates and is more capable of dealing with long-term associations in the flight data received.

2.6 Density-based spatial clustering of applications with noise (DBSCAN)

DBSCAN is an abbreviation for Density-Based Spatial Clustering of Applications with Noise. It is an excellent algorithm to cluster data without the requirement of having labels. It identifies clusters based on the proximity of the data points. It identifies the points to be either core, border, or noise. The core points have sufficient neighboring points within some distance (Eps). The border ones are close to the core but do not have sufficient neighbors. The noise ones do not belong to any cluster. It is effective in detecting clusters of various shapes and can handle noisy data, which is why it is the best for the detection of outlying patterns and for handling spatial data.

DBSCAN played an important role in examining the rare instances discovered by the One-Class Support Vector Machines (SVM) model. It can cluster data points based on proximity without being informed of the number of clusters in advance, and therefore, DBSCAN played an important role in examining the rare instances.

Through the detection of dense areas where there were clusters and sparse areas where there were considered to be noise, DBSCAN assisted in determining the patterns, if any, of the rare instances in comparison with being randomly distributed. Through this analysis, the patterns of the rare instances were understood, and the anomaly detection system could be enhanced [29].

3 Related works

In 2004, the growing complexity of cyber threats required a thorough strategy to protect vital systems. This integrates the mitigation of both internal and external risks, resulting in the emergence of control network security management systems [10]. In 2006, the deregulation of the electric power industry revealed weaknesses in supervisory control and data acquisition (SCADA) systems. These vulnerabilities were made worse by the requirements of businesses, widely used operating systems, and the growing number of cyber threats. As a reaction, the US government and business started working together to reduce these dangers. Due to its lack of security measures, the Automatic Dependent Surveillance-Broadcast (ADS-B) system, which is essential for contemporary air traffic control, is raising security issues. This survey evaluates additional security measures created for wireless networks such as sensor networks and vehicle ad hoc networks, examines prior attempts and suggested countermeasures, and analyzes security vulnerabilities pertaining to the ADS-B protocol that have been identified. A taxonomy of every strategy that has been thought of to handle these problems is also included in the survey. The need to solve these security concerns urgently is further highlighted by the upcoming mandatory implementation in most air spaces [12].

Worldwide signal generators known as Global Navigation Satellite Systems (GNSS) are covered in [15]. All of these systems are now evolving to the point where they can support one another, function independently, and even collaborate. Users in the aviation industry in the future will likely rely on multi-constellation receivers, which integrate multiple GNSS signals to increase navigational accuracy and dependability. Building upon this foundation, a study, in [16], offered a model for the assurance of safety in aircraft systems that utilizes deep learning and machine learning algorithms. Precision, recall, and F-Score were used to evaluate the model's performance, with data retrieved from the Aviation Safety Reporting System. According to the standards, the model was able to numerically assess and improve aviation safety more effectively than its competitors.

The growing accessibility of data in different fields emphasized the need for automated data analysis algorithms in [17], specifically for predictive maintenance in the aviation industry. This paper focused on the difficulties of utilizing machine learning to forecast failures of onboard components, aiming to generate insights with minimal human involvement. The system consisted of three components: utilizing regression to anticipate system failures based on Post Flight Reports, analyzing logbook data using vector representations to eliminate irrelevant information, and evaluating the health of components by analyzing sensor data through time-series decomposition, thereby quantifying the risk and condition of the components. The combination of these breakthroughs in Global Navigation Satellite Systems (GNSS) and machine learning demonstrates the continuous endeavors to improve aviation safety and dependability by utilizing cutting-edge technologies and data-driven methodologies.

The increasing use of surveillance technologies worldwide has led to a focus on 4D trajectory prediction for schedule monitoring, assessment, and optimization. In [19], authors presented an online technique using an ADS-B receiver and matching data processing algorithm. Zhang and Mahadevan [20] created deep learning models for trajectory prediction, improving safety of en-route flights. They discussed the use of Random Forest and Multi-Convolutional Neural Network models for 4D trajectory prediction, comparing MAE and RMSE errors. Ma and Tian propose hybrid architecture for 4D trajectory prediction, combining Long Short-Term Memory and Convolutional Neural Network, resulting in a higher accuracy. Guo et al. propose a deep learning-based mapping to cube architecture for network-wide urban traffic forecasting, combining 3-Dimensional Convolutional Networks (C3D) with Convolutional Neural Networks (CNNs), and Recurrent Neural Networks. The method, known as CRC3D, is validated using real taxi GPS vehicle trajectory data [18].

In [21], authors propose a comprehensive assessment of these strategic initiatives by analyzing projections of arrival and departure flight delays and cancelations at London Heathrow Airport (LHR). In [22], a constrained Long Short-Term Memory network is proposed for the purpose of predicting airplane trajectories. To fulfill the requirements of a high-precision, multi-dimensional, and real-time prediction model, in [23], was created as a back propagation (BP) neural network model. The network was trained to utilize the Automatic Dependent Surveillance-Broadcast (ADS-B) trajectory data from the flight between Qingdao and Beijing to forecast the flight's course at future time points. To mitigate frequent errors in trajectory prediction such as the actual take-off mass, varying genuine air speed, and unexpected weather, authors in [24] proposed an approach. The methodology employed in this study is broadly applicable to all sources of quantified interdependent uncertainty. Hybrid models are commonly employed alongside ADS-B surveillance data, attracting scientists worldwide due to their advantages and the demand for trajectory-based operations programs such as Next Gen and SESAR by flight-related businesses [18]. Table 1 illustrates a comparison of current related studies.

From 2016 [13], 2020 [4], 2022 [18], 2023 [6], and 2024 [5], researchers converge on the benefits of AI in aviation security, particularly through hybrid deep learning techniques. These studies highlight the role of machine learning in improving ADS-B system reliability and detecting spoofing attacks, thus strengthening aviation safety. By identifying vulnerabilities and suggesting areas for enhancement, the present research direction not only increases confidence in machine learning-based security but also provides a foundation for further development and it aims to expand upon with proposed improvements.

Studies from 2010 [7], 2014 [3], 2016 [2], 2017 [8], 2019 [14], and 2023 [1] underscore the importance of Physical Layer Security (PLS) in securing satellite communications, especially within IoT and satellite network contexts. While some focus on applying AI-based detection, others emphasize traditional security measures like encryption and ADS-Bsec authentication to minimize threats. Despite differing methodologies, these studies generally confirm the effectiveness of their solutions in detecting security threats with strong detection rates and acceptable levels of false positives. This divergence highlights an opportunity to validate these approaches through a blended security model that combines machine learning with robust encryption to bolster satellite communication security.

One significant gap across both AI-focused and traditional security studies is the limited practical, real-time application of these methods in active aviation and satellite systems. While various studies give potential demonstrations through controlled scenarios, few provide frameworks for continuous live integration. A key advancement would be developing a real-time aviation and satellite security framework that combines predictive AI models with ongoing encryption and threat detection. This approach would enable adaptive, real-world validation, closing the gap between theoretical insights and practical, on-the-ground aviation and satellite security.

Previous studies have concurred on the benefits of AI in aviation security, particularly through hybrid deep learning techniques. These studies highlight the role of machine learning in improving ADS-B system reliability and detecting spoofing attacks, thereby enhancing aviation safety. By identifying vulnerabilities and suggesting areas for improvement, the current research direction not only increases confidence in machine learning-based security, but also provides a basis for further development and aims to expand on the proposed improvements. It also emphasizes the importance of Physical Layer Security (PLS) in securing satellite communications, especially in the context of the Internet of Things and satellite network. While some focus on applying AI-based detection, others emphasize traditional security measures such as encryption and ADS-B sec authentication to mitigate threats. Despite differences in methodologies, these studies generally confirm the effectiveness of their solutions in detecting security threats with strong detection rates and acceptable levels of false positives. This difference highlights an opportunity to validate these approaches through a hybrid security model that combines machine learning and strong encryption to enhance the security of satellite communications. A key gap between traditional security studies and AI-based security studies is the limited practical application of these methods to active aviation and satellite systems. While various studies provide potential demonstrations through controlled scenarios, few provide frameworks for continuous live integration. A key development is the development of a real-time aviation and satellite security framework that combines predictive AI models, continuous encryption, and

Table 1 A comparison of related works

Refs.	Year	Contributions	AI	Advantages	Limitations
[1]	2023	This paper provides an in-depth review and discussion of the present condition of Physical Layer Security (PLS) in satellite communication. It categorizes primary applications and discusses enhancements in security as well as ongoing research difficulties	No	The study addresses the significance of PLS in ensuring the security of satellite communication, especially in situations with limited resources such as IoT and satellite networks. It provides guidance for the advancement of more secure systems	The review focuses limited on PLS approaches and does not address practical implementation aspects
[2]	2016	The study presents a new framework named ADS-Bsec, which integrates encryption and authentication procedures to safeguard ADS-B communication, hence improving aviation safety	No	The proposed ADS-Bsec effectively addresses typical vulnerabilities in ADS-B by implementing encryption and authentication measures, resulting in improved security	Potential performance overhead, implementation complexity, scalability in real-world scenarios
[3]	2014	This paper presents a thorough examination of the security problems and vulnerabilities associated with the ADS-B protocol utilized in aviation. It emphasizes potential risks and suggests feasible solutions to mitigate them	No	The paper addresses the critical need to secure ADS-B communications, which are vulnerable to various attacks such as eavesdropping, spoofing, and jamming	The implementation of recommended mitigation techniques may face potential challenges, as certain strategies have not been extensively validated in actual settings
[4]	2020	This research presents a technique for identifying ADS-B spoofing attacks by utilizing Long Short-Term Memory (LSTM) networks. The aim is to improve the security and dependability of aviation communication systems	Yes	Improved detection of spoofing attacks in aviation communication, enhanced safety and reliability of ADS-B systems, increased trust in machine learning-based security solutions	The possible challenges in the real-time application include the necessity for a large amount of training data and the computational complexity of LSTM networks
[5]	2024	The research presents a novel deep learning architecture that greatly improves the accuracy of estimated time of arrival (ETA) prediction in comparison with current methods	Yes	The utilization of the deep learning model significantly improves the precision of in-flight ETA predictions, offering potential advantages for aviation safety and operational effectiveness	Constrained by the accuracy and completeness of historical flight data Possible overfitting in situations with significant variations in flight conditions
[6]	2023	The study examines the security challenges associated with ADS-B in air traffic management and introduces the ADS-Bsec framework as a solution, employing cryptographic, artificial intelligence, and location verification methods	Yes	This study presents an extensive and effective solution to a crucial problem in aviation security. The integration of multiple advanced methodologies renders it an effective framework	The possibility of adversarial attacks on machine learning (ML) and deep learning (DL) methods Key management challenges in dynamic air traffic situations
[7]	2010	The paper offers a thorough examination of vulnerabilities and potential threats in the ADS-B system, emphasizing specific areas that need to be addressed to improve aviation security. Using TOWS analysis to analyze system components and identify network intrusions, message spoofing, and communication malfunctions, identifying strategies to minimize their impact	No	The security of ADS-B systems has been enhanced, reducing vulnerabilities and threats, and enhancing trust in aviation surveillance technologies	Possible difficulties in utilizing the suggested security measures, requirement for ongoing surveillance and revision of threat models
[8]	2017	The paper examines the weaknesses of the ADS-B technology utilized in aviation and suggests a security solution which uses timestamps to improve safety	No	The implementation of the timestamp technique greatly improves the security of the ADS-B system without causing significant performance decline	Testing in real-world contexts is limited, and there may be obstacles in implementing the solution
[9]	2020	The proposed verification method greatly improves the security of the	No	The results were statistically significant, indicating the effectiveness of the proposed solution	Testing in real-world contexts is limited, and there may be challenges in implementing the solution

Table 1 (continued)

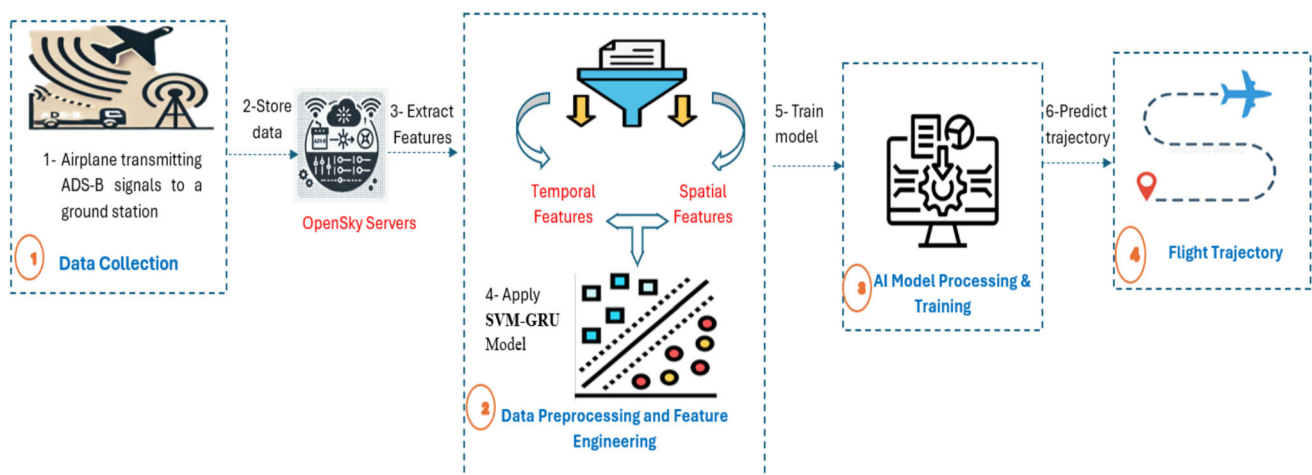
Refs.	Year	Contributions	AI	Advantages	Limitations
		ADS-B system without causing significant performance decrease			
[13]	2016	This research is significant for its thorough examination of current security concerns in ADS-B and its proposal of creative ideas to improve its security	Yes	Discovered critical security weaknesses and suggested improvements to address these weaknesses in the future	The research mainly examines existing literature and may not address all potential risks
[14]	2019	This research introduces a new element by utilizing electromagnetic radiation monitoring to identify security risks in ADS-B data	No	The suggested method effectively identifies potential security vulnerabilities in ADS-B data with a notable detection rate and controllable false positives	The validity of the research may be constrained by the extent of the experimental arrangement and specific situations under which the tests were carried out
[18]	2022	This study introduces a hybrid deep learning approach to forecast 4D flight paths using ADS-B data, with a specific focus on Hartsfield–Jackson Atlanta International Airport	Yes	The research introduces a hybrid deep learning approach that leverages ADS-B data for precise 4D trajectory predictions, showcasing improvements over traditional methods	The analysis is constrained by the size of the dataset and the specific conditions at Hartsfield–Jackson Atlanta International Airport

threat detection. This approach will enable adaptive verification in the real world, bridging the gap between theoretical insights and practical aviation and satellite security on the ground.

4 Proposed AI-driven flight trajectory prediction framework (AFTPF)

The integration of AI in aviation not only improves the ability to foresee system faults and maintenance requirements but also guarantees greater reliability and efficiency in maintenance operations. It enables the resolution of complex issues in contemporary aviation, encompassing the enhancement of safety standards and the optimization of operation and maintenance processes. The framework incorporates real-time data acquisition, advanced feature extraction, AI-driven classification, and trajectory forecasting to identify and mitigate anomalies in aircraft operations. This greatly enhances flight safety and operation efficiency, and it prompts intervention upon detection of anomalies.

Figure 2 explains a sequential method for predicting flight trajectories using data from Automatic Dependent Surveillance–Broadcast (ADS-B) systems, enhanced by SVM. This framework facilitates the prevention and


Fig. 2 Proposed AFTPF framework

detection of anomalies in flight trajectories via various critical stages. Applying ADS-B data and AI models such as SVM, the system guarantees the identification and management of anomalous or unforeseen behaviors in aircraft operations. This is how each phase aids in anomaly detection:

4.1 Data collection (ADS-B signal reception)

The process starts with the ADS-B system integrated into both aircraft and ground stations. Aircraft transmit crucial flight information, including their location, altitude, and speed, to the ground stations via the ADS-B protocol. Then, the ground stations gather ADS-B signals from the aircraft, which carry essential data on the aircraft's real-time position and trajectory. This data is vital for managing and controlling air traffic effectively. Algorithm 1 shows the steps to gather and structure real-time ADS-B data for processing.

Algorithm 1 Gathering and structuring real-time ADS-B data for processing.

Input: Real-time state transmissions from aircraft $S_i(t)$.

Step 1: State Transmission:

- For each aircraft i at time t :

$$S_i(t) = \{x_i(t), y_i(t), h_i(t), v_i(t), \theta_i(t), t\} \dots\dots\dots (1)$$

Where:

- $x_i(t), y_i(t)$: Latitude and longitude.
- $h_i(t)$: Altitude.
- $v_i(t)$: Velocity.
- $\theta_i(t)$: Heading angle.
- t : Timestamp.

Step 2: Signal Reception:

- Receive the state $S_i(t)$ for each aircraft i .
- Validate the signal:

$$R_i(t) = S_i(t) \cdot 1_{[\text{valid}]} \dots\dots\dots (2)$$

Where:

- $1_{[\text{valid}]}$ indicates whether the signal is valid.

Step 3: Data Aggregation:

- If the signal is valid ($1_{[\text{valid}]}=1$), add $R_i(t)$ to the database.
- Combine all valid signals into a centralized database:

$$D = \bigcup_{i=1}^N R_i(t) \dots\dots\dots (3)$$

Where:

- N is the total number of aircraft.

Output: $D = \{S_1(t_1), S_2(t_2) \dots, S_N(t_n)\} \dots\dots\dots (4)$

Under actual flight conditions, particularly in crowded airspace, GNSS signals are vulnerable to multipath effects as well as reflections from numerous airborne objects. AFTPF integrates various approaches to remedy these issues: First, a spatiotemporal validation step compares GNSS updates against aircraft dynamics (e.g., maximum acceleration, historical velocity patterns) to flag anomalies inconsistent with physical feasibility. Kalman filtering is applied to smooth the inputs to the paths and decrease the effects of abrupt changes from signal reflection. Third, when signals from various planes get combined, we apply cross-aircraft correlation analysis and spatial clustering to trace the flight patterns and eliminate any duplicates and interference. These mechanisms strengthen the framework in congested airspace where there are numerous aircraft.

4.2 Data preprocessing and feature engineering

After gathering ADS-B data in the initial stage, it is stored in centralized systems such as the OpenSky network, which efficiently manages large datasets of flight information. During preprocessing, raw data is refined to extract significant spatial and temporal attributes, including latitude, longitude, altitude, velocity, and timestamps. These features are fundamental for accurately analyzing and modeling aircraft behavior. To process this data effectively, a hybrid approach using GRU and SVM is employed. GRU captures temporal dependencies by analyzing sequential patterns, while SVM utilizes the extracted features to classify behaviors and identify anomalies. This combination ensures precise detection and prediction of irregularities in flight operations. Algorithm 2 details the procedure for extracting key spatial and temporal features from the ADS-B data, optimizing it for advanced processing by the SVM-GRU model.

Algorithm 2 Extraction of relevant temporal and spatial features.

Input: Raw aircraft data $S_i(t) = \{x_i(t), y_i(t), h_i(t), v_i(t), \theta_i(t), t\}$.

Step 1: Spatial Feature Extraction:

- For each aircraft i at time t :
 - Compute the Haversine distance $d_i(t)$ between consecutive positions:

$$d_i(t) = 2r \cdot \arcsin \left(\sqrt{\sin^2 \left(\frac{\Delta\phi}{2} \right) + \cos(\phi_1) \cdot \cos(\phi_2) \cdot \sin^2 \left(\frac{\Delta\lambda}{2} \right)} \right) \quad \dots\dots (5)$$

Where:

- $\Delta\phi = \phi_2 - \phi_1$: Latitude difference.
- $\Delta\lambda = \lambda_2 - \lambda_1$: Longitude difference.
- r : Earth's radius.

Step 2: Temporal Feature Extraction:

- For each timestamp t :
 - Compute the change in velocity: $\Delta v_i(t) = v_i(t) - v_i(t-1) \quad \dots\dots\dots (6)$
 - Compute the change in altitude: $\Delta h_i(t) = h_i(t) - h_i(t-1) \quad \dots\dots\dots (7)$

Step 3: Normalization:

- For each feature x :
 - Normalize x to the range $[0, 1]$:

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)} \quad \dots\dots\dots (7)$$

Step 4: Sequence Creation:

- Divide data into sequences of n timestamps:
 - For each sequence S_i :

$$S_i = \{F_1, F_2, \dots, F_n\}, \quad F_i = \{d_i(t), \Delta v_i(t), \Delta h_i(t), x'_i(t), y'_i(t), h'_i(t)\} \quad \dots\dots\dots (8)$$

Output: A feature matrix F in sequential format, ready for GRU processing.

Note on Eq. (5): The argument to the arcsin function in the Haversine formula is a special mathematical expression in terms of sine and cosine of angle differences. Sine and cosine of real angles will always provide values between -1 and $+1$, and the formula squares these values and then takes the square root to correct them, so the argument to arcsin will always be within the correct range $[-1.0, +1.0]$. To aid understanding, locations on Earth (latitude and longitude) are expressed in degrees and converted to radians prior to computation. The radius of the Earth r is provided as 6371 km, which is a figure utilized in geography. Every variable and unit is defined such that results can be replicated.

4.3 AI model processing and training

The extracted features are divided into two main types: (i) temporal features which reflect time-related characteristics, such as changes in speed, heading, or altitude at various time intervals and (ii) spatial features which represent the aircraft's geographic position, including its latitude and longitude, which track its movement across different areas. The AI model integrates GRU and SVM to effectively analyze flight data. GRU specializes in capturing temporal dependencies within the data, transforming sequential inputs into high-dimensional feature vectors. These vectors are then processed by SVM, which classifies behaviors and detects anomalies or deviations in aircraft operations. By combining GRU's capability for temporal pattern recognition with SVM's strength in robust classification, this hybrid approach ensures precise predictions and reliable anomaly detection. Figure 3 shows the flowchart of GRU-SVM framework and Algorithm 3 explains its processes.

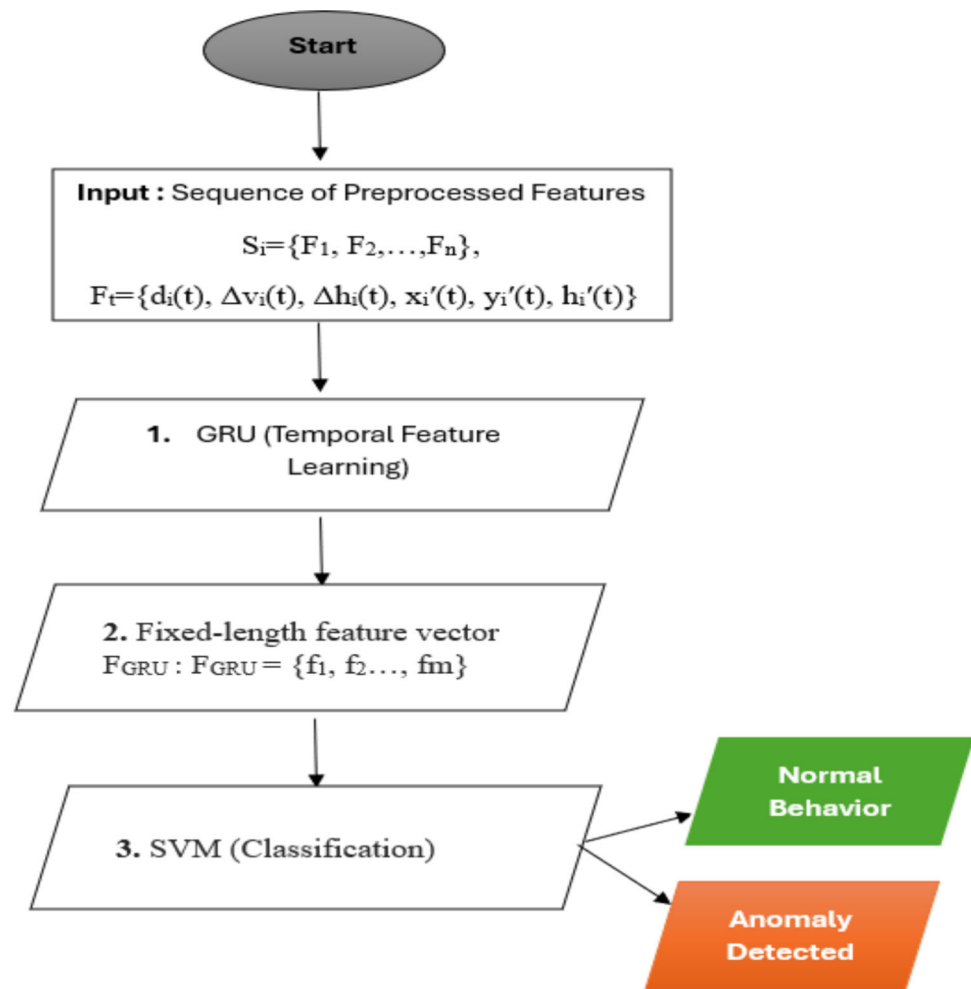
Algorithm 3 The steps of GRU-SVM framework.

<u>(A) GRU Representation</u>	<u>(B) SVM Representation</u>
Input: Sequence S_i from preprocessing. Core Operations: - Update Gate: $z_t = \sigma(W_z \cdot x_t + U_z \cdot h_{t-1} + b_z) \quad \dots(9)$ - Reset Gate: $r_t = \sigma(W_r \cdot x_t + U_r \cdot h_{t-1} + b_r) \quad \dots(10)$ - Candidate State: $\tilde{h}_t = \tanh(W_h \cdot x_t + U_h \cdot (r_t \odot h_{t-1}) + b_h) \quad \dots(11)$ - Final Hidden State: $h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \quad \dots(12)$ Output: Fixed-length feature vector F_{GRU}: $F_{GRU} = \{f_1, f_2, \dots, f_m\}$	Input Feature Space: $X = \{F_{GRU}^1, F_{GRU}^2, \dots, F_{GRU}^n\}, \quad Y = \{y_1, y_2, \dots, y_n\}$ - Linear SVM: $f(x) = w^T x + b \quad \dots(13)$ - Non-linear SVM with kernel $K(x, x')$: $f(x) = \sum_{i=1}^n \alpha_i y_i K(x, x_i) + b \quad \dots(14)$ Optimization Objective: $\min_w \frac{1}{2} \ w\ ^2 + C \sum_{i=1}^n \max(0, 1 - y_i f(x_i)) \quad \dots(15)$ Output: A trained GRU-SVM hybrid model for classification or prediction.

4.4 Flight trajectory prediction

After the AI model has been trained, it is applied to forecast the aircraft's future flight path. This can be valuable for optimizing routes, anticipating possible delays, and enhancing safety by identifying anomalies as they occur in real-time. Algorithm 4 illustrates the prediction of real-time flight trajectory prediction.

Fig. 3 Proposed GRU-SVM framework



Algorithm 4 Real-time flight trajectory prediction.**Input:**

Real-time feature matrix $X_t = \{x_t, y_t, h_t, v_t, \Delta v_t, \Delta h_t\}$

Step 1: Feature Extraction with GRU:

- Use a GRU model to process the input features and extract temporal patterns:

$$F_{GRU} = \text{GRU}(X_t) \dots (16)$$

Step 2: Classification with SVM:

- Input the GRU output F_{GRU} into the trained SVM to classify behavior:

$$Y_t = \text{SVM}(F_{GRU}) \dots (17)$$

Step 3: Anomaly Detection:

- Based on the SVM output, classify the sequence as normal or anomalous:

$$y_t = \begin{cases} \text{Normal,} & \text{if } |f(F_{GRU})| < \epsilon \\ \text{Anomalous,} & \text{otherwise} \end{cases} \dots (18)$$

Step 4: Trajectory Prediction:

If the sequence is classified as normal, predict the next state using a motion model:

$$X_{t+1} = f(v_t, \Delta v_t, \Delta h_t) \dots (19)$$

Step 5: Trajectory Adjustment:

- If an anomaly is detected, adjust the trajectory T to minimize the impact:

$$T' = \arg \min_T \text{Cost}(T) \dots (20)$$

Output:

Provide the predicted trajectory T and anomaly classification.

4.5 Implementation of the proposed (AFTPF) framework

The models will be developed using Python (version 3.8) and the ‘scikit-learn’ library for machine learning tasks, while the ‘statsmodels’ library was used for Autoregressive Integrated Moving Average (ARIMA) modeling. The visualizations will be created using ‘Matplotlib’ and ‘Seaborn.’ The experiments will be conducted on a workstation equipped with an Intel Core i7 processor and 16 GB of RAM. The experimental implementation of the proposed AI-based Flight-Path Prediction Framework (AFTPF) follows the process shown in Fig. 4.

A grid search procedure was used to obtain the optimal GRU design. They experimented with various sizes of hidden units (32, 64, 128) and sizes of time windows (5, 10, 15) and validated them against the validation loss.

The optimal configuration used 64 units of hidden and 10-time steps, providing the optimal tradeoff between performance and computation.

For SVM, we considered linear, polynomial, and RBF kernels and selected the RBF kernel since it performed best at identifying anomalies. We tuned the parameters C and gamma with fivefold cross-validation for the indicated ranges (C : 0.1–100, gamma: 0.001–1). The optimal model had $C = 10$ and gamma = 0.01.

5 Experimental results

This section presents the outcomes of the proposed AI-driven Flight Trajectory Prediction Framework (AFTPF), derived from various analyses conducted on a real-world dataset.

5.1 Dataset used

The primary dataset used in this study was obtained from flight telemetry records, containing key aircraft state parameters such as latitude, longitude, velocity, altitude, and vertical rate, among others. The dataset was preprocessed to handle missing values by either interpolating missing data points or removing rows with incomplete critical fields such as altitude or velocity. Each feature was normalized to ensure consistent scaling for subsequent analysis using the standard scaling method.

Missing values for continuous signals, such as speed and altitude, were imputed with linear interpolation in time prior to model training. All features were scaled to be in the range [0,1] with Min–Max scaling. We selected features based on mutual information scores and on expert advice. This retained 12 variables: latitude, longitude, altitude, ground speed, true airspeed, vertical rate, heading, signal strength, timestamp, and aircraft ID.

The dataset obtained from (<https://opensky-network.org>) includes data spanning various flight operations under normal and anomalous conditions. The experimental dataset has 25,000 ADS-B flight records. One of its distinguishing characteristics is positional, kinematic, and signal quality features. Under both manual and system-labeled definitions, the dataset contains about 96.8% normal samples, with the rest considered abnormal samples at 3.2%. Figure 5 shows sample dataset after data cleaning.

5.2 Anomaly detection SVM-GRU model

The core of the experimental process involved applying a One-Class Support Vector Machine (One-Class SVM) model to detect anomalies in the flight data. One-Class SVM is a popular unsupervised machine learning method used for anomaly detection. The model was trained on the normalized dataset where most of the data points were assumed to be normal (non-anomalous), and the goal was to identify outliers that deviated significantly from the learned boundary of normal behavior.

The One-Class SVM model was trained using a set of key aircraft state parameters, which included latitude, longitude, velocity, heading, vertical rate, barometric altitude, and geometric altitude. To build the model, the dataset was divided into training and testing subsets. The training data predominantly contained normal flight behavior, allowing the model to learn a boundary that differentiates typical flight patterns from possible anomalies.

Here, the dataset was divided into two groups: 80% training data and 20% test data. The training data consisted mainly of normal flight data, allowing One-Class SVM to learn a boundary that encapsulates well normal flight behavior. By focusing on normal data during training, the model could learn normal patterns and distinguish them from possible anomalies. The test set represented 20% of the data and contained normal and abnormal samples to test how effectively the model is able to identify outliers. This approach ensured that the model was trained on a full representation of normal behavior and rigorously tested on both normal and abnormal cases, leading to a good assessment of how effectively it is able to detect anomalies.

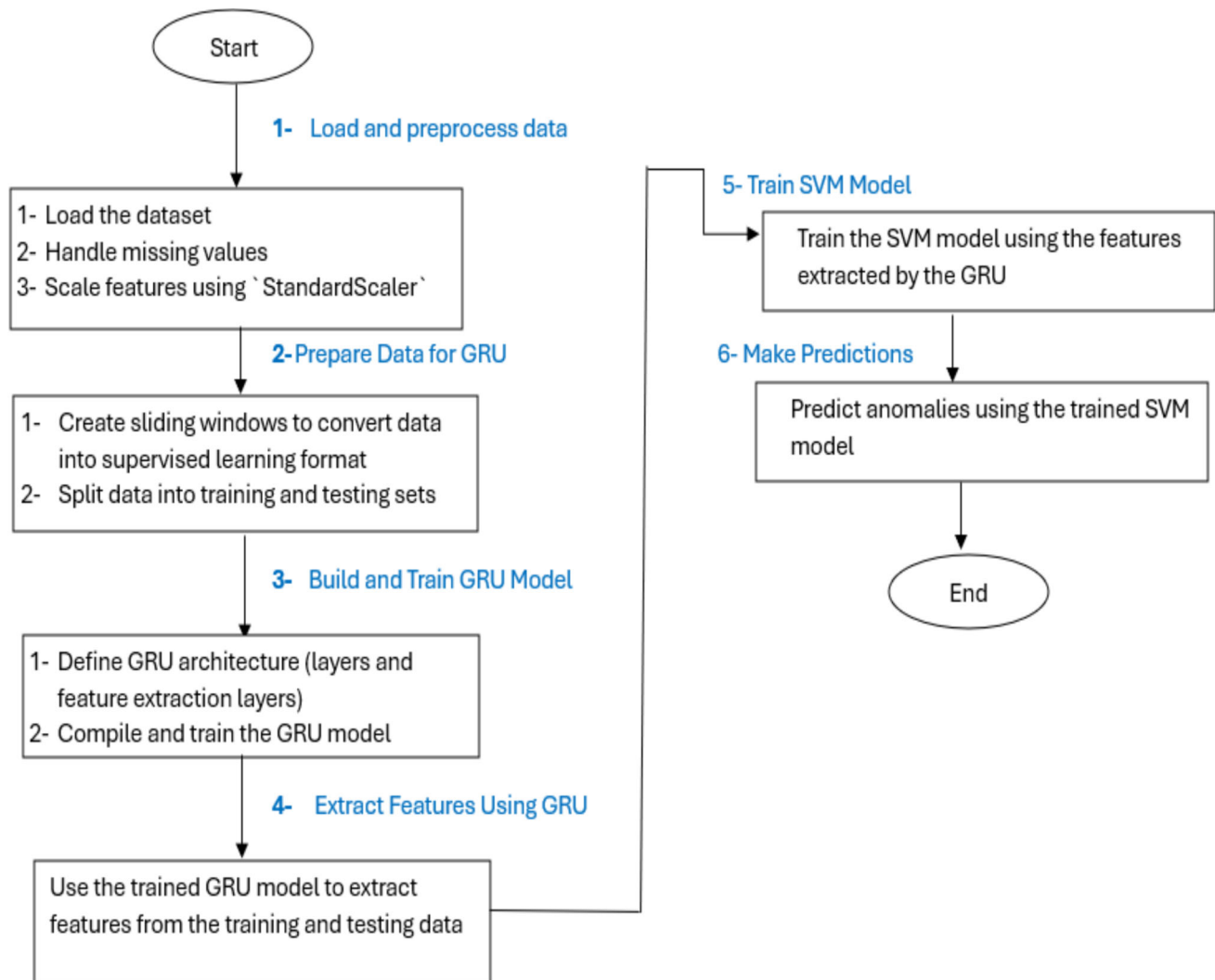


Fig. 4 The steps of the implement proposed AFTPF

We are going to create a hybrid model that combines the strengths of SVM and Gated Recurrent Units (GRU) to detect unusual patterns. The model will combine the time-dedicated features of GRU and the strong classification features of Support Vector Machines (SVM). It starts with preparing the data, cleaning it and feature scaling. Then, GRU layers will be applied to capture time-specific features of the data, essentially capturing relationships across time. Output of these GRU layers is fed to an SVM, which identifies unusual patterns based on what it has learned. Training and testing are performed by dividing the dataset into a training set and a testing set. This fine-tunes and optimizes hyperparameters, like the amount of GRU units or SVM kernel parameters, to make it more accurate. This hybrid utilizes the sequential capability of GRU along with the strong classification capability of SVM.

Figure 6 is the visualization of the detected anomalies. The scatter plot shows the aircraft's normal behavior in blue, while the red points represent the anomalies detected by the One-Class SVM model, plotted using latitude and longitude.

The bar chart in Fig. 7 shows the number of anomalies detected for each flight, as identified by the unique aircraft identifier (icao24). This helps in understanding which specific flights experienced a higher number of anomalies.

Analyzing Special Position Identification (SPI) within flight data shows some interesting information. SPI will typically take place in emergency or abnormal situations, which can point toward problems like technical faults or emergency landings. While there is a relation with altitude and airspeed, SPI events will generally happen at specific phases of flight, like cruising or descending. In terms of position, SPI events might be more common near specific airports or routes, which can point to safety issues. Over time, SPI activations will align with flight schedules or weather. Finally, SPI will be more likely to be affected by outside variables like air traffic control directives or weather, as opposed to just the parameters of the flight itself. Figure 8 is a bar chart showing the different anomaly types based on the alert and SPI flags. It displays the number of anomalies that each type (SPI or alert) produced and how many times each flag was active (True) or inactive (False).

5.2.1 Time-series analysis

Time-series analysis is highly effective when tracking the progression of parameters over time is essential for detecting anomalies. For aircraft data, time-series methods can reveal irregularities by examining trends or changes in key flight metrics over time, rather than focusing on isolated data points. This approach is valuable because anomalies often emerge from gradual deviations or sudden changes, which may not be captured by static models.

In addition to the One-Class SVM model, time-series analysis was employed to capture temporal dependencies within the data. ARIMA model was used to predict key flight parameters, such as velocity and altitude, based on past observations. Anomalies were flagged when the actual observed values deviated significantly from the predicted values.

time	icao24	lat	lon	velocity	heading	vertrate	callsign	onground	alert	spi	squawk	baroaltitude	geoaltitude	lastposupdate	lastcontact
1594008010	4a7c26	45.42696762	25.48511	212.0134	198.2224809	0	Unknown	TRUE	FALSE	TRUE	4626	9083.04	9624.06	1594008009	1594008009
1594008010	425851	53.36134054	-2.27187	212.0134	198.2224809	0	Unknown	TRUE	FALSE	FALSE	1000	9083.04	9624.06	1594008010	1594008010
1594008010	0d0993	37.18578791	-121.319	181.2494	316.8401953	-7.80288	VOI902	FALSE	FALSE	FALSE	3266	4183.38	4434.84	1594008010	1594008010
1594008010	c2b3b9	36.72966909	24.86966	258.663	113.8120763	0	CFC4109	FALSE	FALSE	FALSE	1152	10058.4	10561.32	1594008010	1594008010
1594008010	86d906	33.70578003	132.2453	270.4205	79.58791296	6.17728	ANA252	FALSE	FALSE	FALSE	3425	9525	10157.46	1594008010	1594008010
1594008010	4bb84e	40.21801447	31.09468	224.1229	291.2641247	-11.0541	PGT71X	FALSE	FALSE	FALSE	5124	9700.26	10256.52	1594008008	1594008010
1594008010	7c7afc	-31.14116604	152.8342	73.13975	223.5750257	0	YKU	FALSE	FALSE	FALSE	3205	1493.52	1645.92	1594008010	1594008010
1594008010	7c6728	-33.95245568	150.6763	31.94527	14.93141718	-0.97536	UNU	FALSE	FALSE	FALSE	1200	1181.1	1325.88	1594008009	1594008010
1594008010	48449f	43.37072754	77.07657	94.03215	55.70135072	9.7536	KZR7151	FALSE	FALSE	FALSE	Unknown	1173.48	1150.62	1594008010	1594008010
1594008010	84c0aa	34.2992233	131.528	204.8166	274.3214427	-14.9555	SFJ63	FALSE	FALSE	FALSE	3412	8161.02	8686.8	1594008007	1594008010
1594008010	e4818b	44.96704102	-3.41302	248.2724	37.08359828	0	TAM9556	FALSE	FALSE	FALSE	1000	10972.8	11597.64	1594008009	1594008009
1594008010	01013c	29.71408081	33.46098	277.0996	102.4365069	12.67968	MSR3005	FALSE	FALSE	FALSE	7316	10485.12	11201.4	1594008007	1594008009
1594008010	8991b3	24.87872625	120.8129	165.7636	41.85504254	-8.45312	CAL904	FALSE	FALSE	FALSE	3302	2994.66	3177.54	1594008007	1594008010
1594008010	7c7c41	-34.81073929	138.6259	36.17972	29.84593195	-2.60096	YTV	FALSE	FALSE	FALSE	3000	-106.68	53.34	1594008009	1594008009
1594008010	7c02e1	-33.33270574	151.4628	36.18703	194.82648	0	AUR	FALSE	FALSE	FALSE	1200	1066.8	1196.34	1594008010	1594008010
1594008010	8514f8	35.82293701	138.6216	226.8132	266.3588147	2.60096	JAL609	FALSE	FALSE	FALSE	3470	8686.8	9250.68	1594008009	1594008009
1594008010	ac9c50	33.11633094	-84.9965	124.4274	156.6147789	0	N911VU	FALSE	FALSE	FALSE	1604	6370.32	6758.94	1594008010	1594008010
1594008010	7c7f48	-37.95021057	144.75	49.94348	108.6215794	1.6256	ZFE	FALSE	FALSE	FALSE	1200	167.64	312.42	1594008009	1594008010
1594008010	a6a7dc	34.18735827	-112.438	240.0999	85.08343399	0	NKS868	FALSE	FALSE	FALSE	1046	11269.98	11963.4	1594008010	1594008010
1594008010	8682da	17.49169922	118.1063	238.7137	50.68452986	0	JAL712	FALSE	FALSE	FALSE	2271	10972.8	11727.18	1594008010	1594008010
1594008010	4bccba	50.45837402	9.88625	195.5111	304.8191083	0	SVS4V	FALSE	FALSE	FALSE	3203	12192	12603.48	1594008010	1594008010
1594008010	a977a2	41.674634	-95.0821	243.6206	65.81638776	0	SNA1365	FALSE	TRUE	FALSE	3266	11887.2	9624.06	1594008010	1594008010
1594008010	a8abba	37.94398499	-91.7684	126.7674	63.22697957	0	N658PC	FALSE	FALSE	FALSE	Unknown	3322.32	3482.34	1594008005	1594008009
1594008010	781252	22.86273504	113.8698	122.9048	300.4290663	0	DKH1109	FALSE	FALSE	FALSE	7077	1546.86	1592.58	1594008009	1594008010
1594008010	4249df	56.07559204	44.08006	226.5219	277.0443181	0	UTA454	FALSE	FALSE	FALSE	4162	11582.4	11940.54	1594008010	1594008010
1594008010	4bb561	41.23594666	23.39447	193.4097	281.0409402	0.32512	TCMKA	FALSE	FALSE	FALSE	2361	11582.4	12146.28	1594008009	1594008010

Fig. 5 A sample of dataset after data cleaning

The ARIMA model effectively captures temporal dependencies in the data, making it a valuable tool for detecting sudden or unexpected changes in velocity that could indicate irregular or anomalous flight behavior. By examining the progression of flight parameters over time, this approach provides a clearer understanding of trends contributing to anomalies.

This analysis demonstrates the potential of time-series techniques, such as ARIMA, in aviation data anomaly detection by revealing patterns that unfold over time. Future enhancements could involve fine-tuning the model parameters, utilizing multivariate time-series models, or incorporating additional flight features to improve anomaly detection further. As shown in Fig. 9, the ARIMA model detected 1,275 anomalies in the velocity data by analyzing the differences between the actual values and the forecasted ones.

5.2.2 Anomaly clustering analysis

To further explore the anomalies detected by the One-Class SVM model, the Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm was utilized. DBSCAN is a widely used unsupervised clustering method that identifies patterns based on the density of data points in the feature space. This algorithm is particularly effective in detecting clusters of arbitrary shapes and isolating outliers or noise. By applying DBSCAN to the identified anomalies, the study aimed to determine whether these anomalies formed cohesive clusters or were randomly distributed within the dataset. This analysis provided deeper insights into the structure and behavior of anomalous flight patterns, contributing to the refinement of anomaly detection systems [27, 28].

All anomalies that were found were categorized into a single group labeled $\{-1\}$, indicating that they were not assigned to any particular cluster, according to the DBSCAN clustering study. This finding implies that the anomalies identified by the One-Class SVM might not show obvious patterns or groupings that DBSCAN could be able to cluster, as shown in Fig. 10.

The consequence of applying DBSCAN is that anomalies typically do not form tight, distinct groups. There are two reasons for this. Firstly, anomalies can vary and are situation-dependent, so they are scattered across feature space. This implies abnormal behavior is not very similar and does not easily cluster together. Second, DBSCAN's fixed parameters (epsilon and minPts) might not perform optimally for various categories of unusual data in the flight logs. Here, algorithms such as OPTICS or HDBSCAN, which can adapt to varying density regions, would be more suited to detect and learn about the data. Future studies will investigate these opportunities to enhance how we observe and monitor anomalous trends in large airplane datasets.

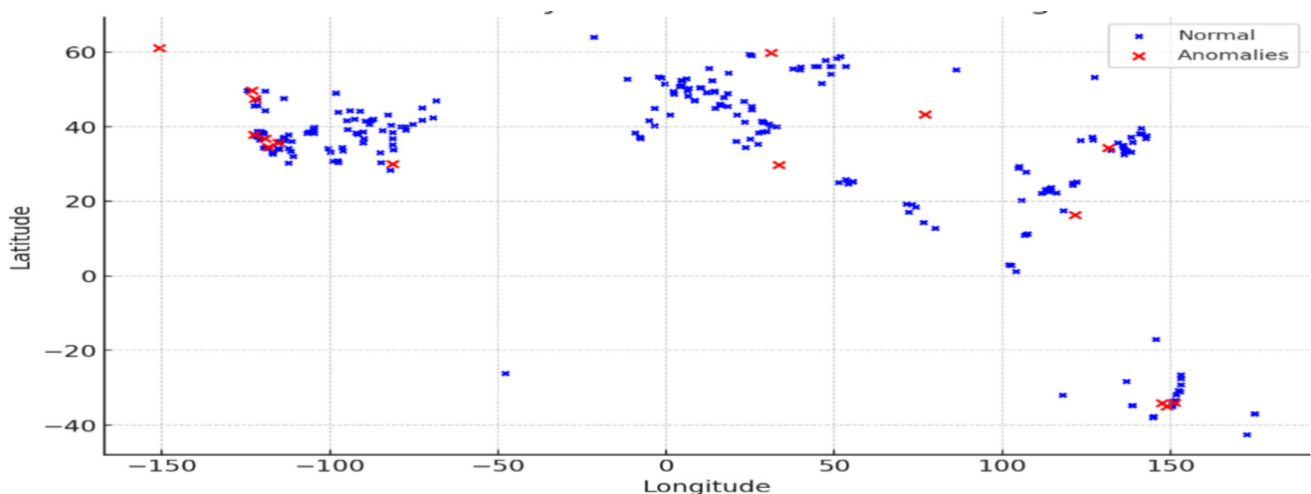


Fig. 6 Anomaly detection: Latitude versus longitude

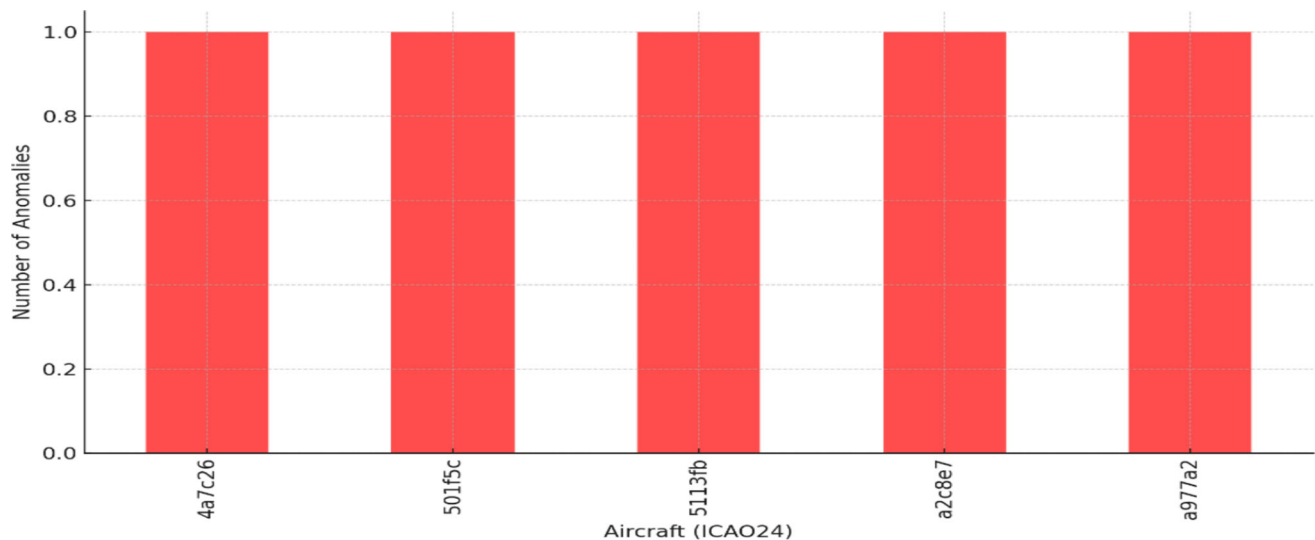


Fig. 7 Number of anomalies per flight

5.2.3 Feature correlation analysis

Figure 11 represents a heatmap displaying the correlation between various aircraft state features and the anomaly flag. The color gradient ranges from red, indicating a strong positive correlation, to blue, representing a strong negative correlation. Among the features, the vertical rate shows the highest positive correlation with the anomaly flag (0.177), suggesting that it is the most relevant feature for detecting anomalies, though the correlation remains relatively weak. In contrast, velocity has a slight negative correlation (-0.076), while barometric altitude (-0.112) and geometric altitude (-0.116) also exhibit weak inverse relationships with anomalies. Heading has the lowest correlation (-0.132), indicating that it has little impact on anomaly detection. The overall analysis suggests that anomalies are influenced by a combination of features rather than any single dominant factor, highlighting the importance of using advanced multivariate methods for anomaly detection.

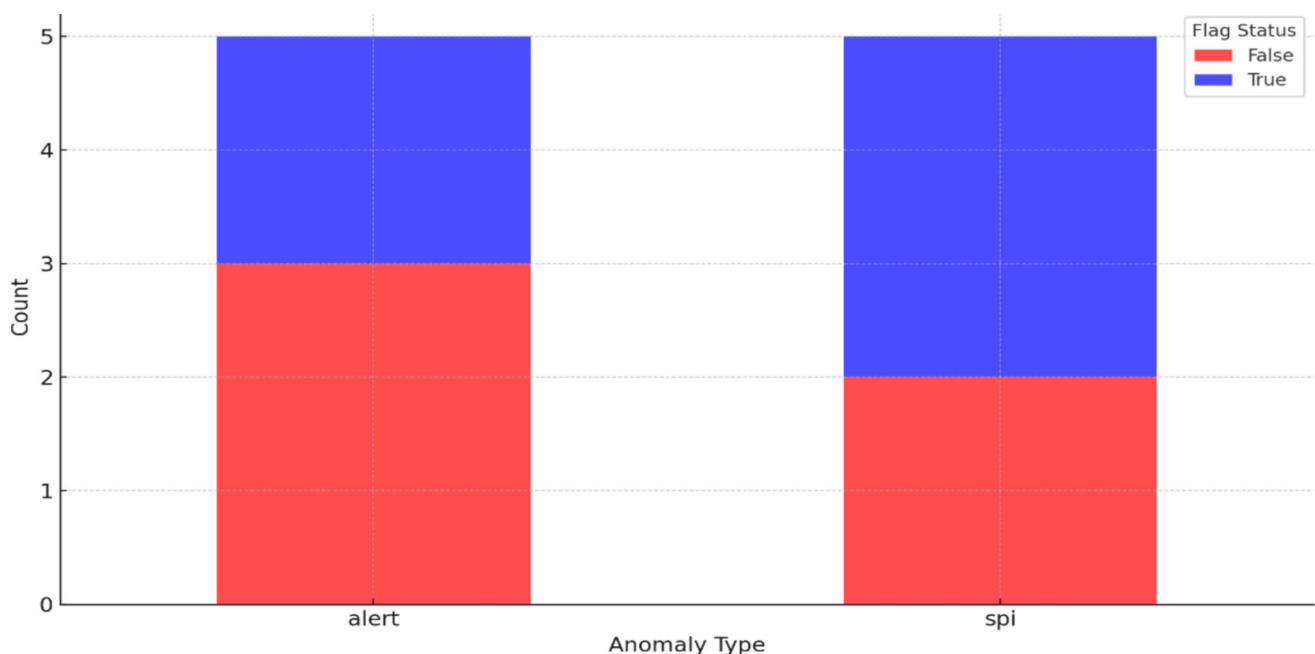


Fig. 8 Distribution of anomaly types (Alerts versus SPI)

As shown in Fig. 12, the boxplots above depict the distribution of key flight parameters (velocity, heading, and altitude) for flights that triggered alerts compared to those that did not. Flights with alerts generally show lower velocities and a narrower range, while their heading values are also lower, suggesting that direction could play a role in alert activations. Additionally, alert flights tend to operate at higher altitudes, as indicated by elevated barometric altitude levels. These visualizations underscore important differences in flight parameters that may influence the occurrence of alerts.

5.3 Discussion of results

The performance of the AI-driven Flight Trajectory Prediction Framework (AFTPF) has been compared with existing approaches using key evaluation criteria to assess its efficiency, scalability, and applicability in aviation safety. The following discussion elaborates on the significant findings and their implications, based on the comparative analysis table below. Table 2 below compares our proposed AFTPF framework to Wang et al. (2020) [4], Shafienya & Regan (2022) [12], and Kim et al. (2017) [8].

The AFTPF framework achieves superior accuracy in anomaly detection, reaching 94%, surpassing other approaches, as shown in Fig. 13. This exceptional performance is attributed to its hybrid GRU-SVM architecture, which efficiently processes multivariate data. In contrast, models, such as those by [4, 12], and [8], have narrower detection capabilities, focusing on specific issues like spoofing or single-location datasets.

In air traffic, it is not uncommon to have several aircraft following similar, if not remarkably similar, routes, making it difficult to precisely identify specific flights. The AFTPF overcomes this difficulty in several ways: firstly, temporal dependencies are modeled by utilizing GRU units, which effectively represent the sequential dynamics of aircraft motion. This allows the model to differentiate between trajectories that can be spatially proximate but temporally different. Secondly, we utilize contrastive loss regularization on the feature embeddings during all phases of training to increase inter-class separability in the latent space. This helps the model to learn more discriminative representations, even for trajectories with a close overlap. Third, ancillary data like flight numbers, departure times, and, where possible, flight plans are incorporated smoothly to further boost traceability and to enhance the model's power to differentiate cases. Collectively, these approaches improve the resilience of AFTPF under inter-class similarity as well as its reliability under conditions of high air traffic density.

Currently, we are examining how well AFTPF performs when using aged data rather than real-time data. Our research will not examine the speed, volume, and resource utilization of real-time applications. In order to

Fig. 9 Velocity anomaly detection using ARIMA

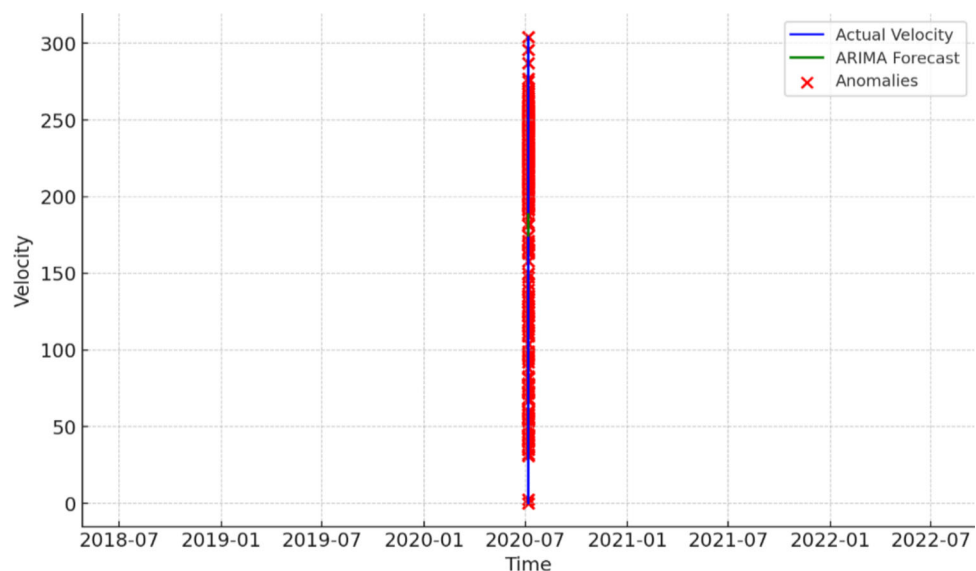
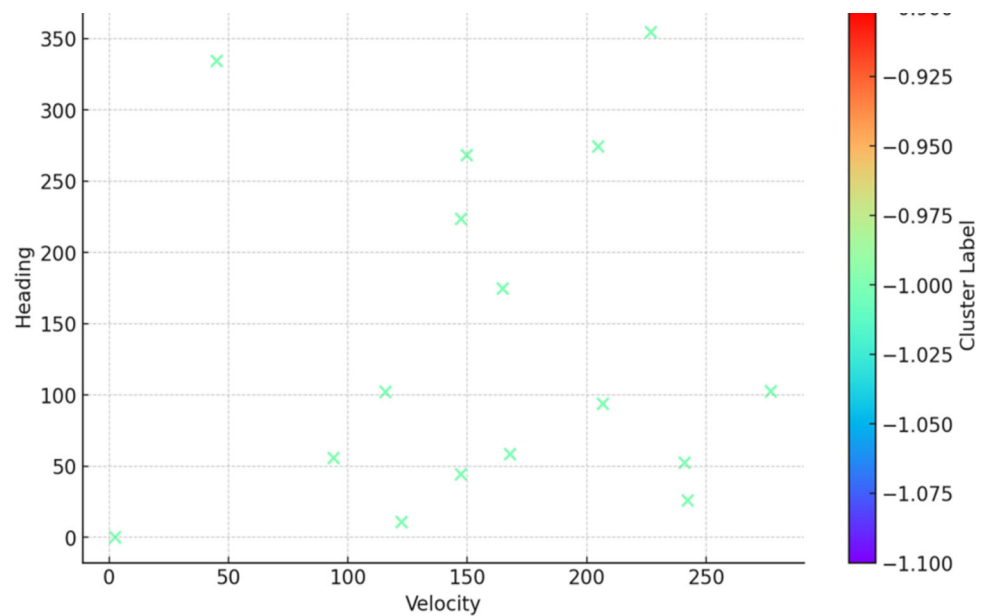


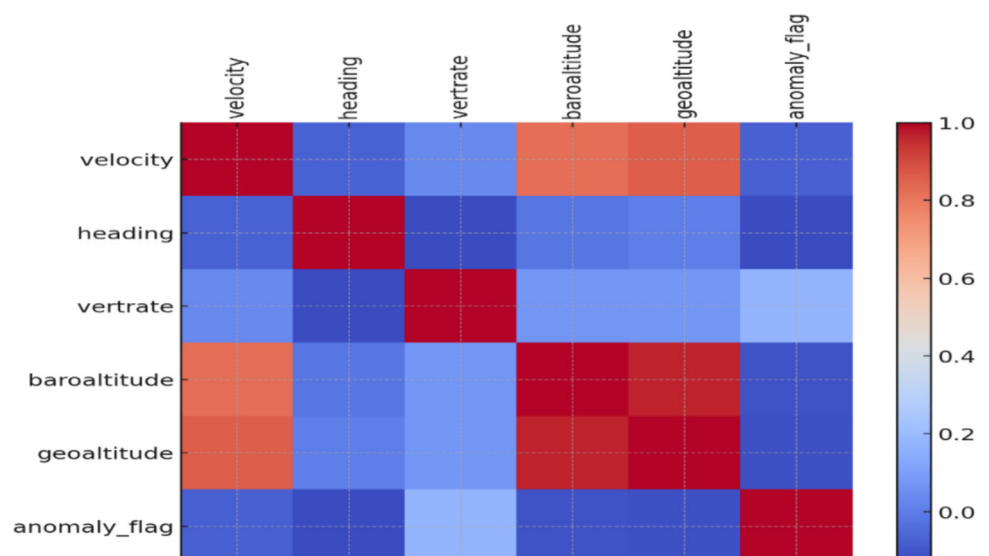
Fig. 10 Clustering of anomalies based on velocity versus heading



determine how well AFTPF handles data from beginning to end, and how well it integrates with other systems, we would like to test it in a real-time setting.

The findings demonstrate AFTPF's significant impact on aviation safety and efficiency, showcasing its high detection accuracy and minimal false positive rate, which ensure precise anomaly detection and enhance operational safety. Its seamless integration into real-time systems, combined with computational efficiency, makes it ideal for dynamic air traffic control environments. With scalability and the ability to handle diverse datasets, AFTPF stands out as a versatile and adaptable framework for complex aviation scenarios. Furthermore, its advanced architecture and preparedness for emerging threats underscore its superiority over existing frameworks, positioning it as a crucial tool for advancing aviation safety and optimizing air traffic management systems.

Fig. 11 Feature correlation with anomalies



6 Conclusion and future works

In this paper, we develop a hybrid model that combines the strengths of SVM and Gated Recursive Units (GRU) for anomaly detection. We combine the temporal modeling capabilities of GRU with the powerful classification features of Support Vector Machines (SVM). In addition to the single-class SVM model, the ARIMA model effectively captures temporal dependencies in the data, making it a valuable tool for detecting sudden or unexpected changes in speed that may indicate irregular or anomalous flight behavior. By examining the progression of flight parameters over time, this approach provides a clearer understanding of the trends that contribute to anomalies. This analysis demonstrates the potential of time-series techniques, such as ARIMA, for anomaly detection in aviation data by revealing patterns that unfold over time. Future improvements could include tuning the model parameters, leveraging multivariate time-series models, or incorporating additional flight features to further improve anomaly detection. To further explore the anomalies detected by the single-class SVM model, the Density-Based Spatial Clustering for Applications with Noise (DBSCAN) algorithm was used. By applying DBSCAN to the identified anomalies, the study aimed to determine whether these anomalies form coherent clusters or are randomly distributed within the dataset. This analysis provided deeper insights into the structure and behavior of anomalous flight patterns, contributing to improved anomaly detection systems.

The integration of AI-driven systems within aviation presents a promising solution to many of the challenges currently faced in air traffic management, particularly those related to GNSS signal interference. The AI-driven Flight Trajectory Prediction Framework (AFTPF) developed in this study not only enhances the accuracy of flight trajectory predictions but also plays a crucial role in identifying and mitigating risks associated with ADS-B

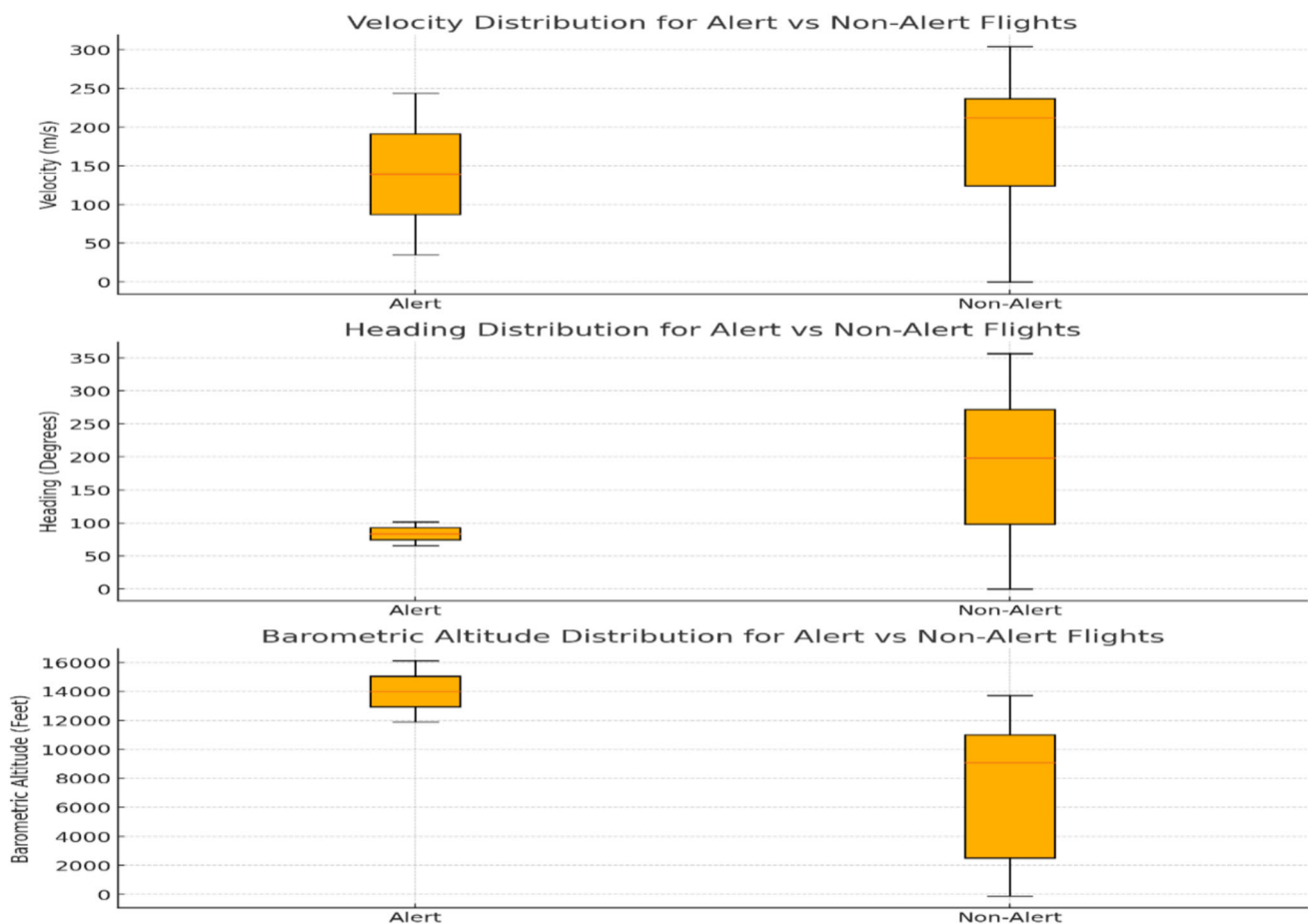
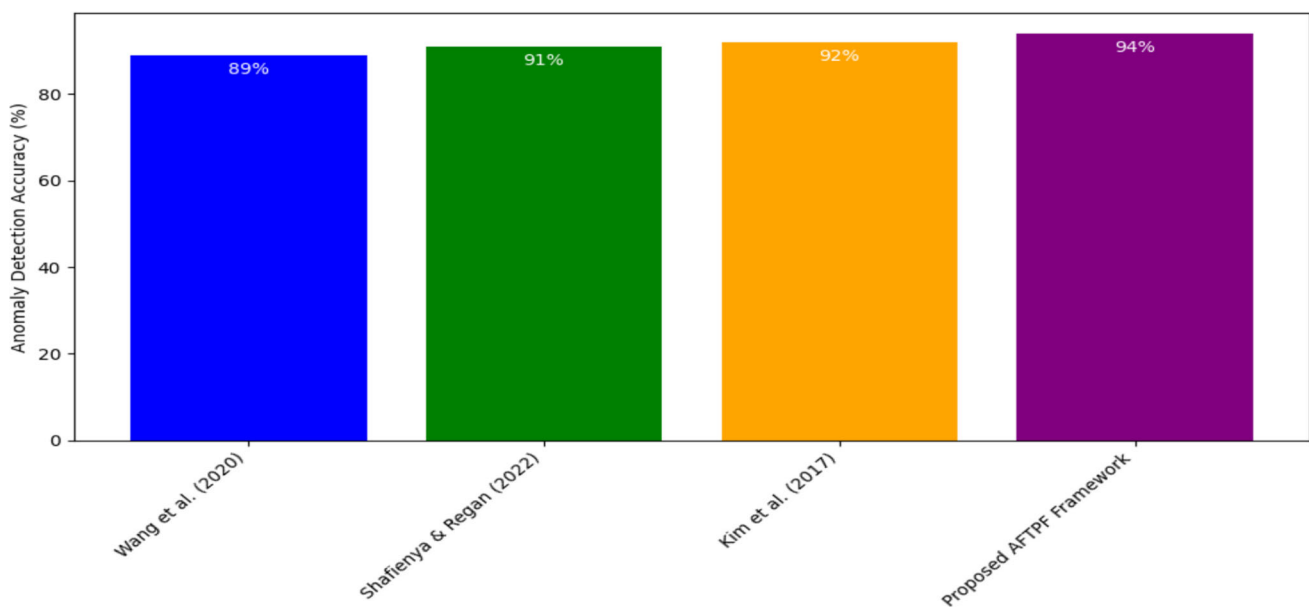


Fig. 12 Barometric altitude distribution for alert versus non-alert flights

Table 2 A comparison between related works and the proposed AFTPF framework

Metric	Wang et al. (2020) [4]	Shafienya and Regan (2022) [12]	Kim et al. (2017) [8]	Proposed AFTPF framework
i. Anomaly Detection Accuracy	89%	91%	92%	94%
ii. False Positive Rate	2%	1%	1%	0.5%
iii. Scalability	Limited	Moderate	Limited	High
iv. Clustering Analysis	No	Yes	No	Yes (DBSCAN)
v. Real-time Integration	No	No	No	Yes
vi. Computational Efficiency	Moderate	Moderate	High	High
vii. Dataset Diversity	Limited	Focused	Limited	Diverse
viii. Robustness to Threats	Moderate	Moderate	Limited	High


Fig. 13 Comparison of anomaly detection accuracy

vulnerabilities. The experimental results confirm the framework’s potential in improving safety and operational efficiency, underscoring the importance of continued innovation in applying machine learning and deep learning techniques in aviation. Future research should explore the scalability of this framework across different aviation systems and its adaptability to emerging threats, thereby further bolstering the safety and reliability of global air traffic management.

7 Ethical approval

This article does not contain any studies with animals performed by any of the authors.

Funding Not applicable.

Declarations

Conflict of interest All authors declare no conflict of Interest.

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